

➤ **Topic:** *Feature-Based Failure Analysis of Grok Against Weak Opponents*

➤ **Dataset:** 358 losses across 6 languages where Grok win-rate < 35%

❑ Objective

Study Goals

- Identify models where **Grok underperforms** (win-rate < 35%).
- Extract **response-level features** from Grok's losses.
- Compare Grok vs. opponent behavior across:
 - Length
 - Sentence structure
 - Emojis & formatting
 - Code usage
 - Conversation metadata
- Identify **language patterns** and **failure trends**.

❑ Dataset Overview

Evaluation Dataset

- Total matches with Grok involved: **N = 1358**
- Total Grok losses (after adding both_bad as losses): **358**
- Weak opponents = models with Grok win-rate < 35%
- Losses analyzed only against these weak opponents

❑ Weak Opponent Identification

Filtering Logic

- Opponent = the model on the other side of the match
- Grok win = grok_in_a & winner=model_a or grok_in_b & winner=model_b
- Loss = anything not win/tie **plus** both_bad
- Weak Opponents = win-rate < 35%

opponent	matches	wins	ties	losses	win_rate(%)
chatgpt-4o-latest-20250326	155	54	30	54	34.8
deepseek-r1-0528	123	42	17	50	34.1
gemini-2.5-pro	171	42	34	75	24.6
gemini-2.5-pro-preview-03-25	44	13	10	17	29.5
gemini-2.5-pro-preview-05-06	49	11	8	25	22.4
o3-2025-04-16	158	50	31	54	31.6

❑ Loss Distribution by Language

Languages with ≥ 10 Grok Losses

- **en:** 194 losses
- **pl:** 26 losses
- **ru:** 22
- **und:** 30
- **zh:** 18
- **de:** 13

→ **All further analysis focuses only on these languages.**

❑ Key Feature Definitions

Features Extracted:

Response Content

- Character length
- Word length
- Sentence count
- Emoji count
- Code-block count
- Punctuation use

Conversation Context

- Conversation length (characters)
- Number of turns

Metadata

- Metadata length
- Emoji presence in metadata

Relative Metrics

- Grok/Opponent length ratio (chars & words)

❑ Global Summary (358 Losses)

Global Averages

- Grok response len: **4250 chars, 598 words**
- Opponent response len: **3476 chars, 471 words**
- Grok sentences: **47.5**
- Opp sentences: **41.4**
- Emojis (Grok/Opp): **0.21 vs 0.92**
- Code blocks: **1.38 vs 1.44**
- Length ratio (chars): **1.61**
- Length ratio (words): **1.59**

Interpretation

- Grok tends to **over-explain** compared to opponent.
- Opponents use more emojis and slightly more code.
- Grok's punctuation is higher → more verbose writing.
- Higher length ratios mean Grok writes **significantly longer** than the opponent when losing.

❑ Per-Language Comparison (Means)

Key findings:

- **EN:** Grok very verbose — 4830 vs 3643 chars
- **RU / PL:** similar trend: Grok consistently longer
- **ZH:** Opponents write longer than Grok
- **DE:** Almost equal
- **UND:** close but opponent slightly longer in sentences & emojis

❑ Interpretation of Language Effects

Insights

- Grok tends to lose in languages where it **over-generates** text.
- In Chinese (zh), opponents produce longer and clearer text → Grok underperforms due to **under-explanation**, not over-explanation.
- High emoji use by opponents suggests some models follow a **friendlier / less formal style**, while Grok stays formal.
- German (de) losses seem content-driven, not stylistic.

❑ Style Mismatch Analysis

Observations

- Grok losses strongly correlate with:
 - **Excessive length**
 - **High punctuation density**
 - **Low emoji usage**
 - **More sentences, but lower clarity**

Opponent Advantage Patterns

- Opponents often write:
 - Shorter but **denser** content
 - More structured answers
 - Slightly friendlier tone (emojis)
 - More direct answers with fewer filler lines

❑ Metadata Observation

Metadata Results

- Metadata length stable: ~460 chars
- No emojis in metadata (as expected)
- Metadata not a differentiator
 - Grok's failures come from *response content*, not metadata or prompt structure.

❑ Key Failure Signatures

Across 358 Grok losses

- 1.Overwriting (too long)
- 2.Too many sentences → lower clarity
- 3.Lower usage of emojis & friendliness markers
- 4.Underuse of code blocks in technical contexts
- 5.Opponent responses more concise and structured
- 6.Mismatch with expected format (lists, structure)

❑ Most Critical Finding

Weak-opponents = **models where Grok had <35% win-rate**

But those models were actually:

- More structured
- More concise
- More style-aligned with human preferences on those prompts

**The term “weak-opponents” is misleading — they exploit Grok’s weaknesses.*

❑ Future Work

Next Steps

- Extract deeper formatting features:
 - Markdown headers
 - List depth
 - Code quality scoring
 - Table detection (extended TSV parser)
- Detect:
 - Tone (formal / casual)
 - Toxicity
 - Readability score (Flesch, etc.)
- Classify prompts into categories (math, coding, translation...)